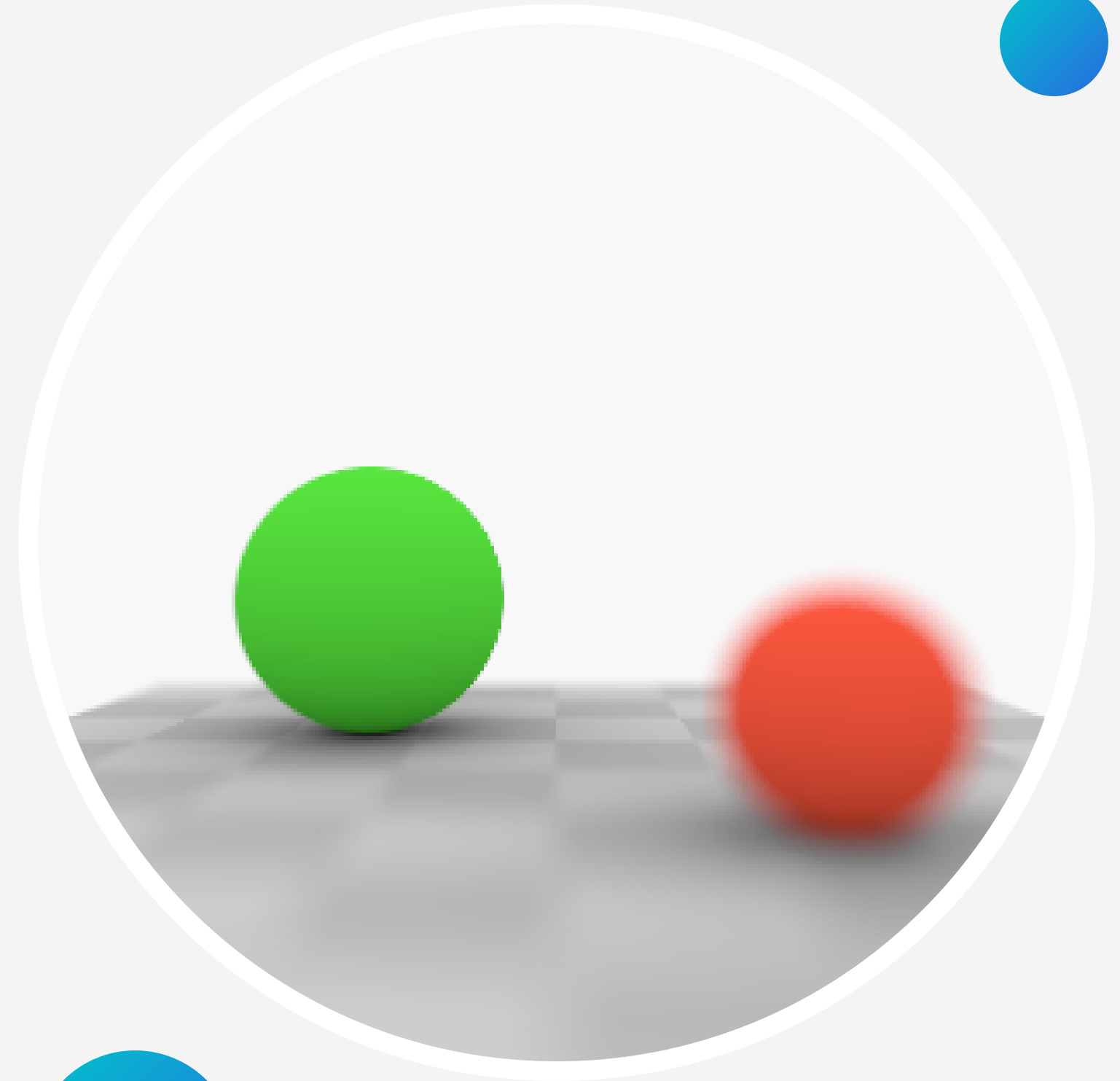


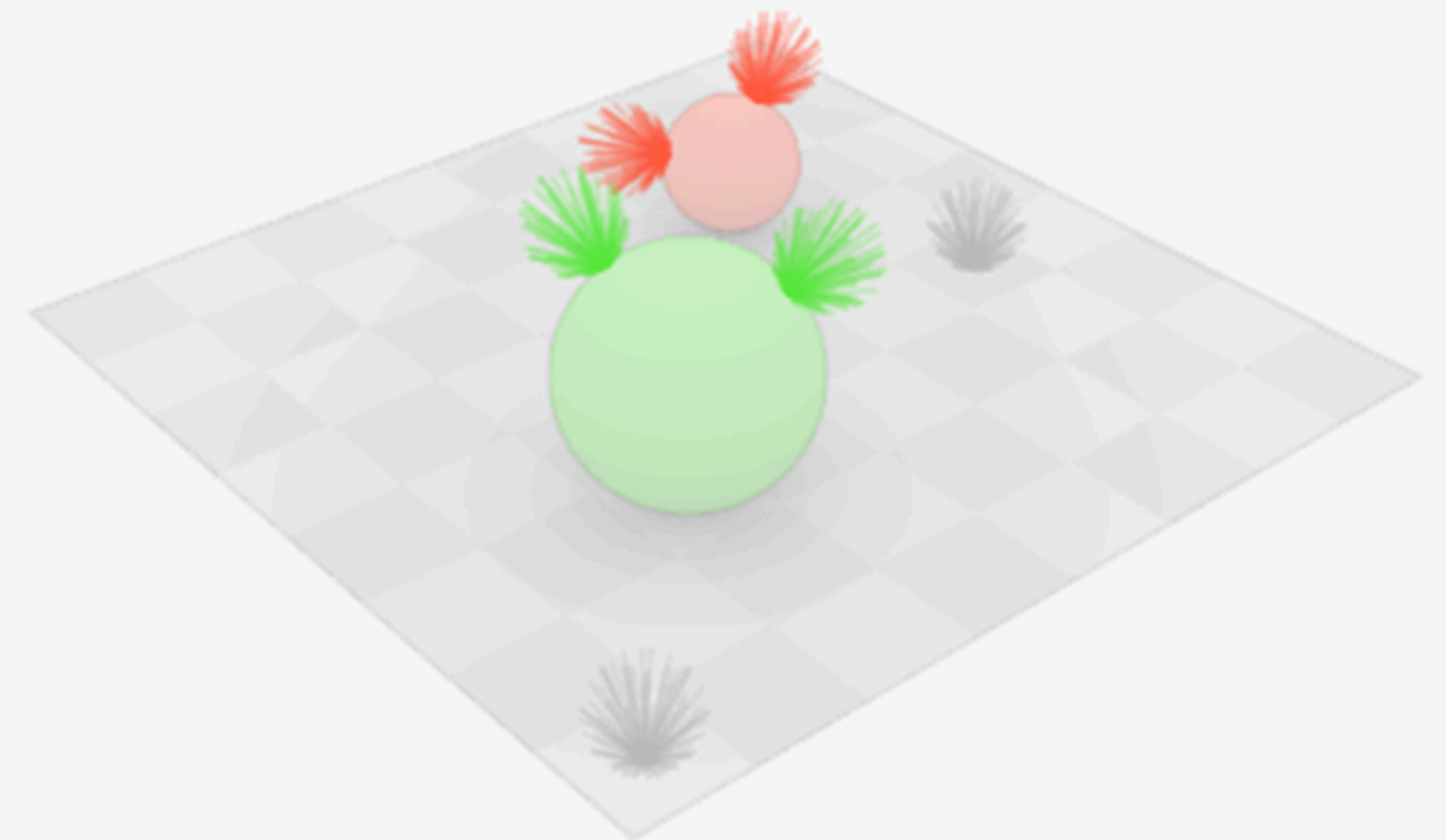
LAB2 TUTORIAL

CAMERAS AND LENSES



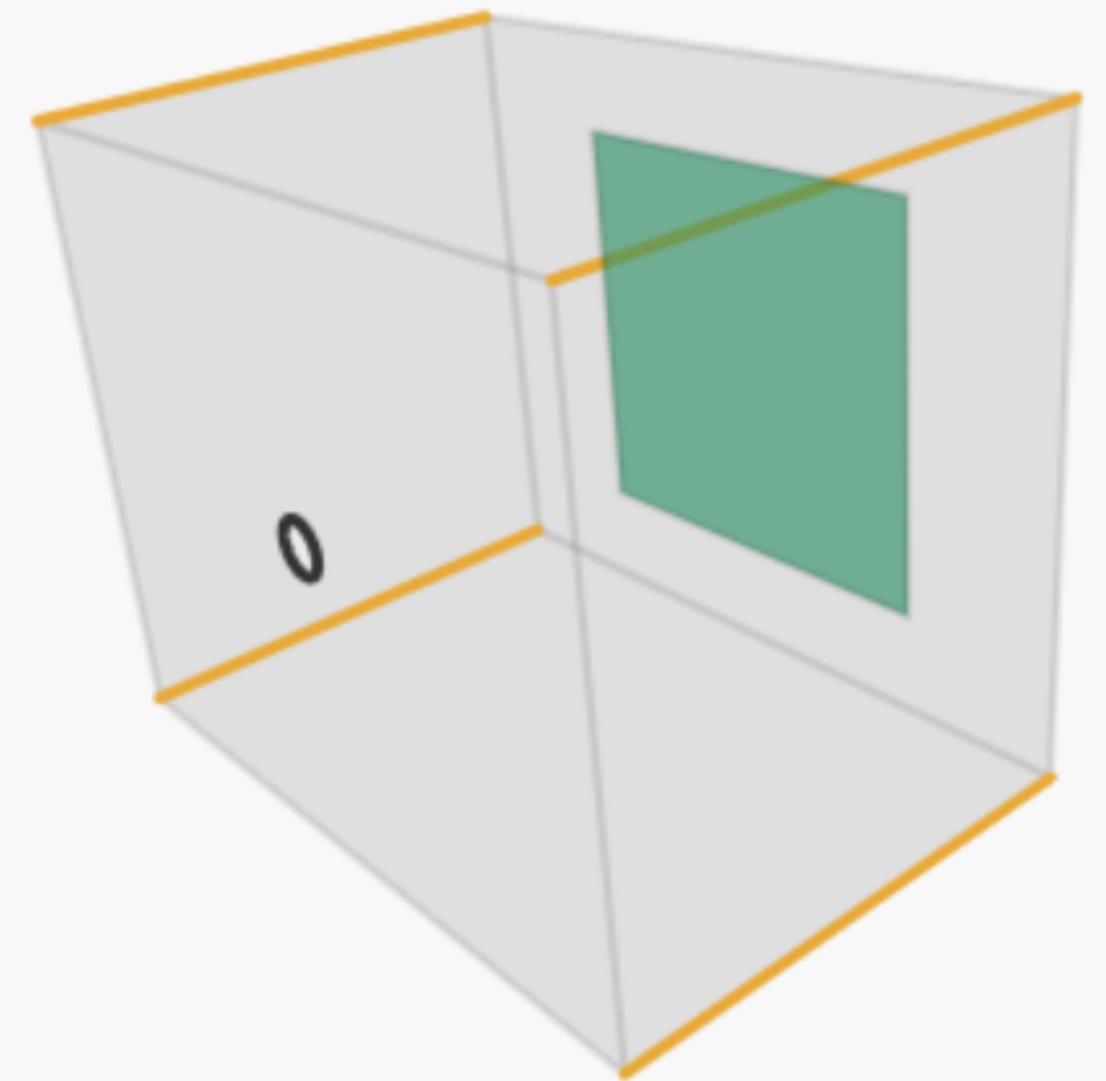
PINHOLE CAMERA IMAGING PRINCIPLES

- Unlimited Light Exposure Problem:
 - Without a physical limitation, every pixel on the sensor is exposed to light from the entire environment.
 - This results in a completely chaotic image with no meaningful structure.



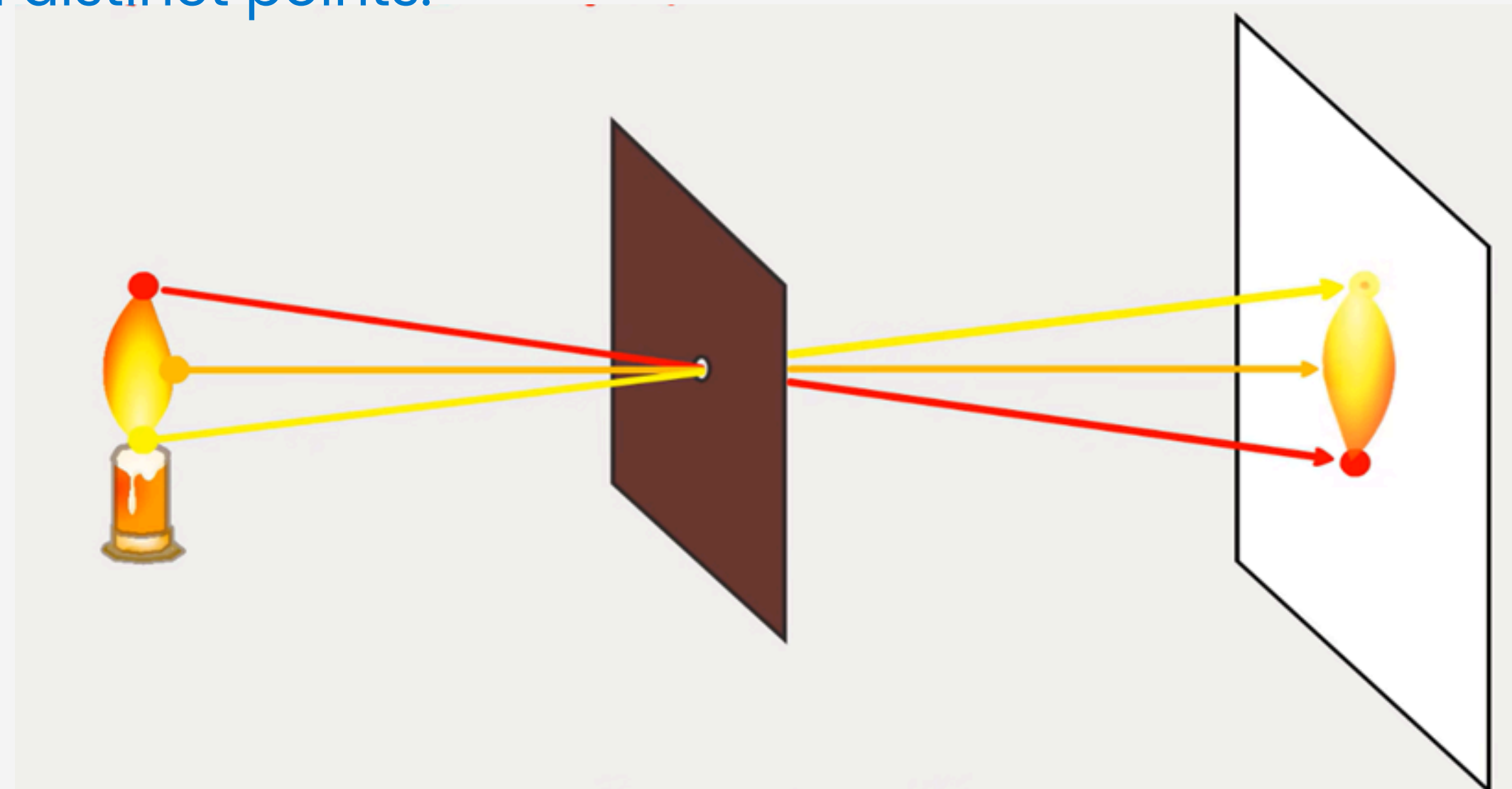
RESTRICTING LIGHT DIRECTION

- Placing a box with a small hole in front of the sensor can restrict the direction of incoming light, allowing only light from specific directions to reach particular pixels.



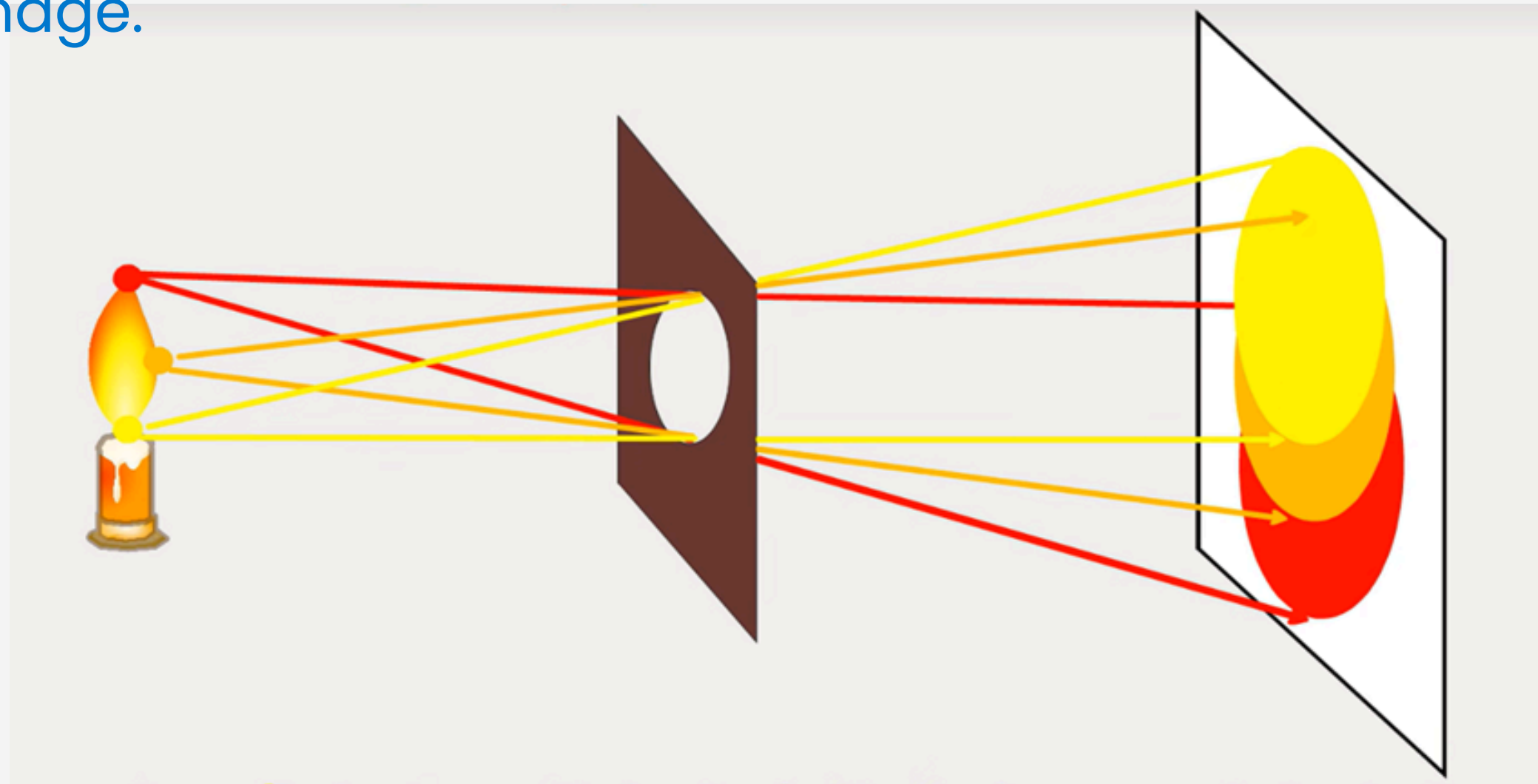
FUNCTION OF THE PINHOLE

- Light rays from the object pass through the small pinhole and converge on the sensor.
- Each ray forms a small light spot on the screen, reflecting the object's features.
- The orderly arrangement of these spots reconstructs the complete image of the object.
- The image is inverted due to the crossing of light rays (180° rotation).
- The focused light forms sharp and distinct points.



FUNCTION OF THE PINHOLE

- A larger pinhole causes light from a single point to spread, forming larger light spots.
- As a result, the light spots formed by different points on the object overlap multiple times on the screen.
- Consequently, any given light spot on the screen no longer corresponds meaningfully to the original light source, nor does it reflect the characteristics of the emitting object, making it impossible to form a clear image.



SENSOR TO PINHOLE DISTANCE

- Effect of Changing the Distance Between the Pinhole and the Sensor might be:
- Field of View (FoV)
 - Increasing the Distance, the field of view becomes narrower, and objects in the image appear larger.
- Sharpness:
 - When the Distance is Too Large, the image may become blurry
 - Because the light cone allowed through the pinhole becomes larger. Light rays from different points may spread out on the sensor, reducing sharpness and clarity.

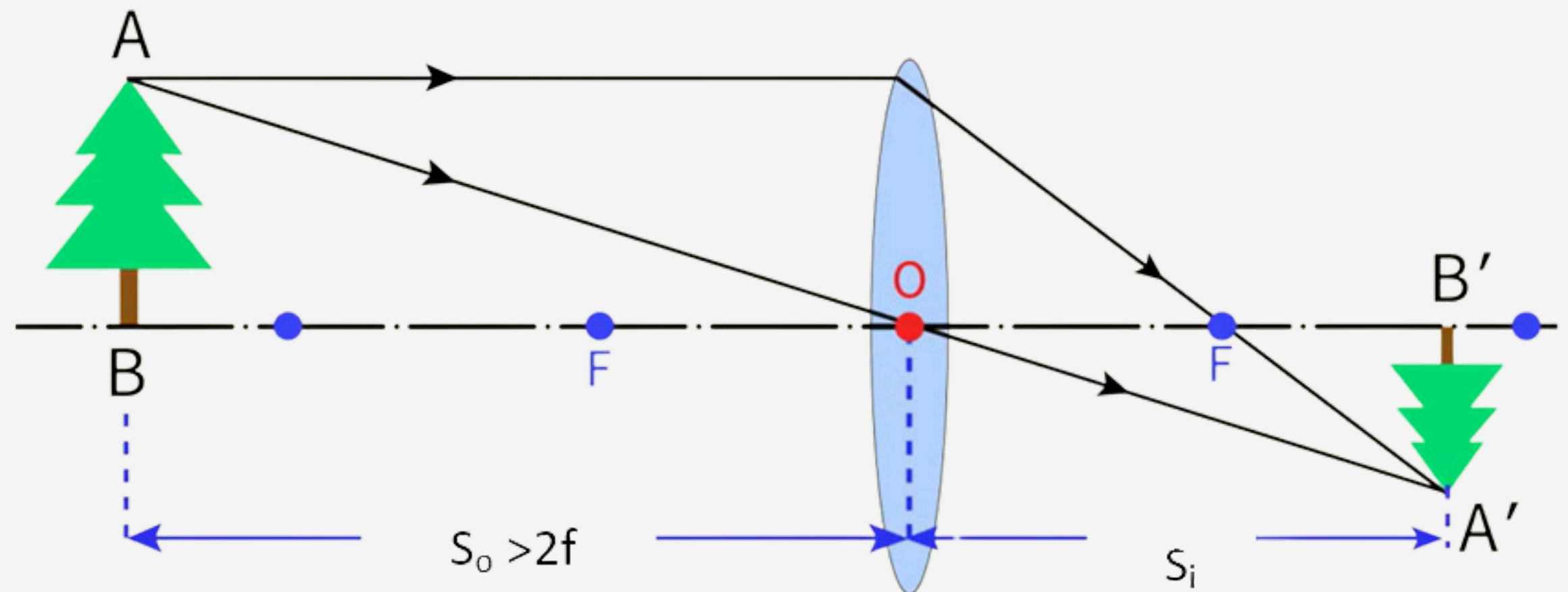
PRINCIPLES OF IMAGE FORMATION IN CAMERAS

- The image formed by a camera depends on the relationship between the object distance and the focal length.
- What Happens when object distance $> 2f$:
 - The image is inverted, real, and smaller than the object.
- As the object moves closer to the lens, the image enlarges.
- When object distance $= 2f$, the image size equals the object size.

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$$

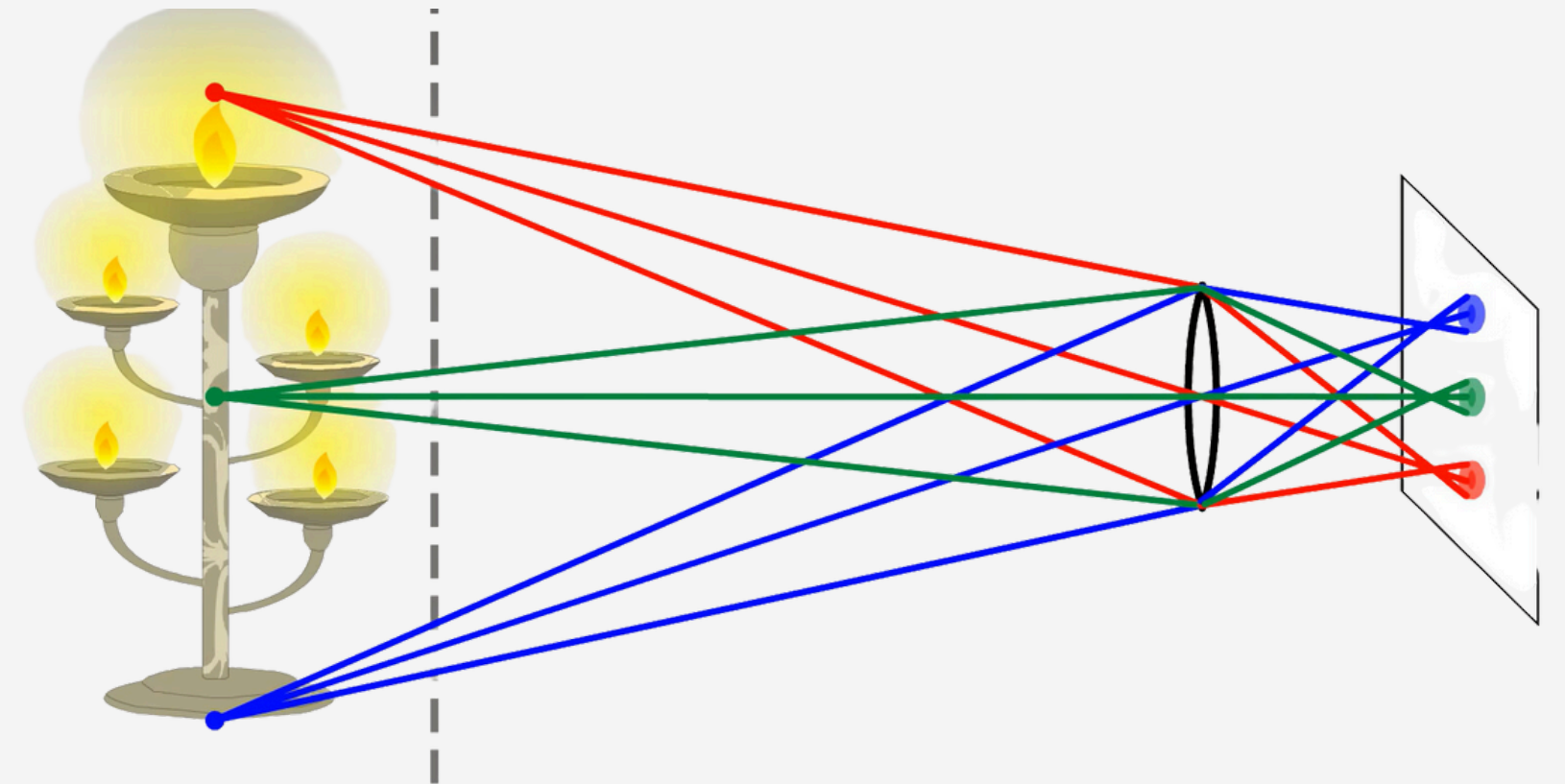
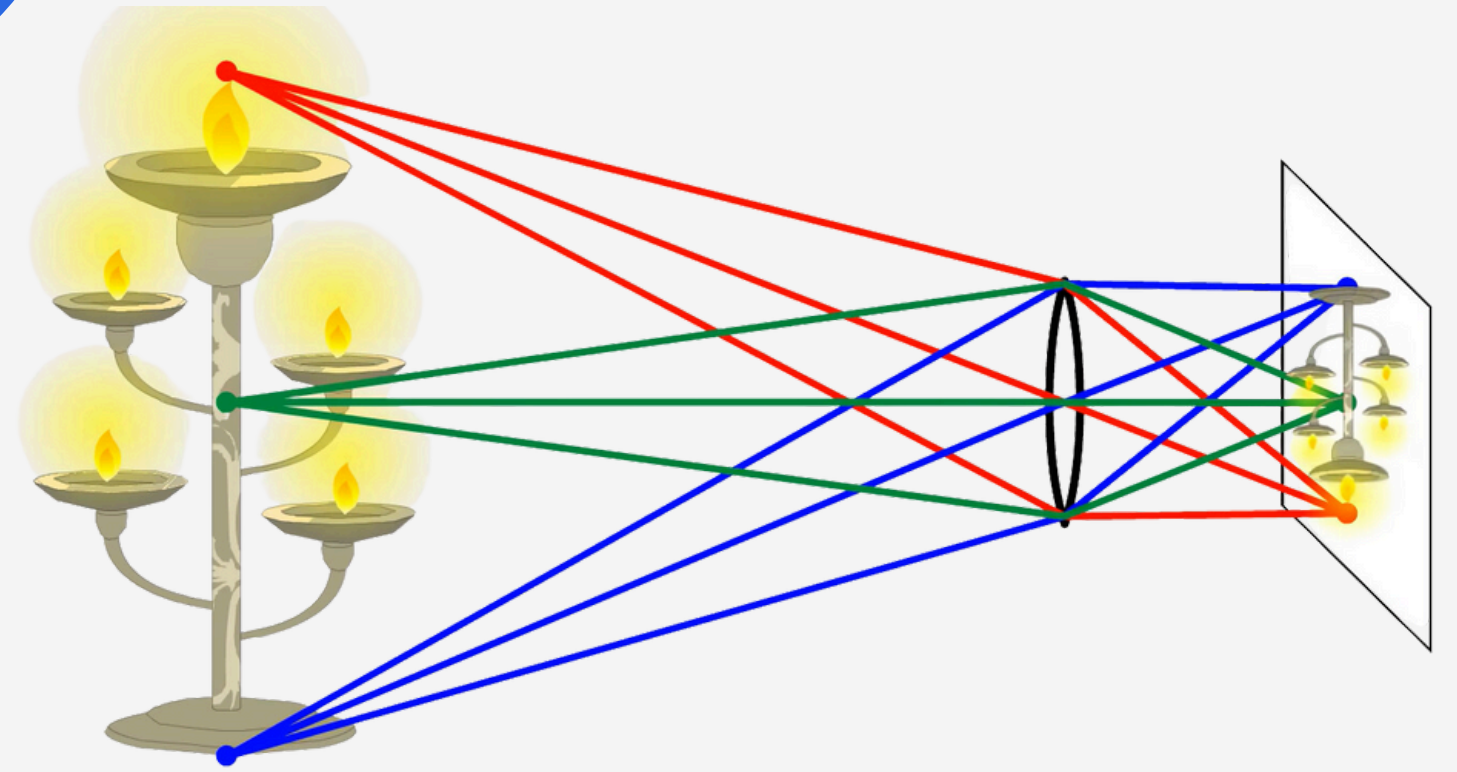
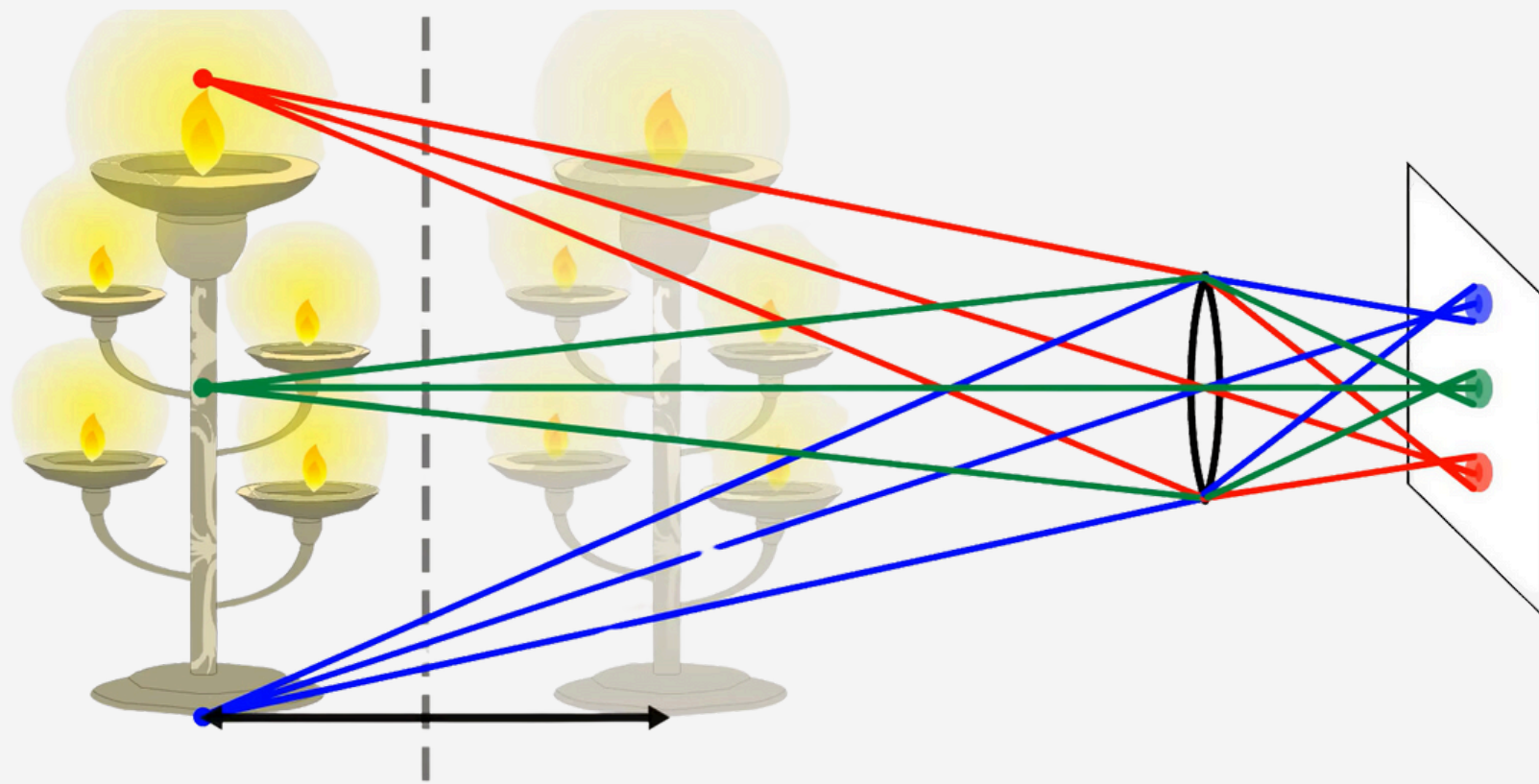
$$M = \frac{s_i}{s_o}$$

- f : Focal length
- M : Magnification



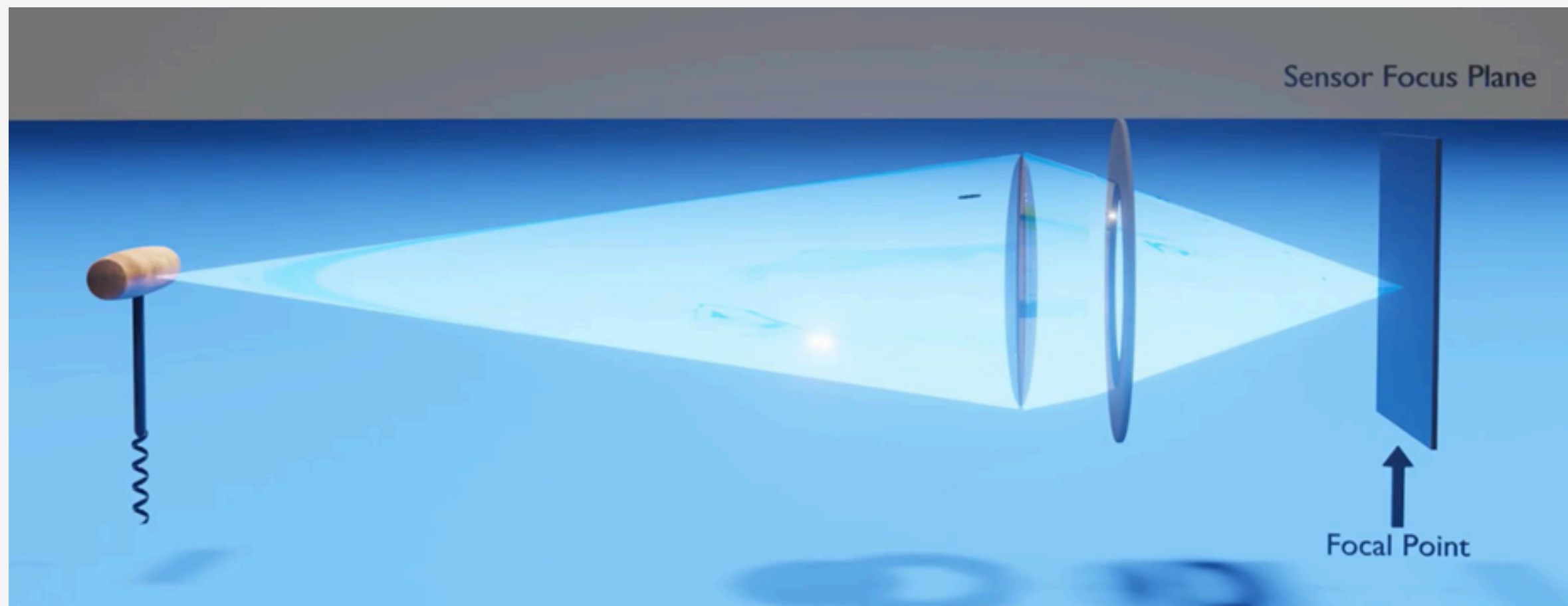
DEPTH OF FIELD

- Depth of Field (DOF) refers to the range in an image where objects appear acceptably sharp.



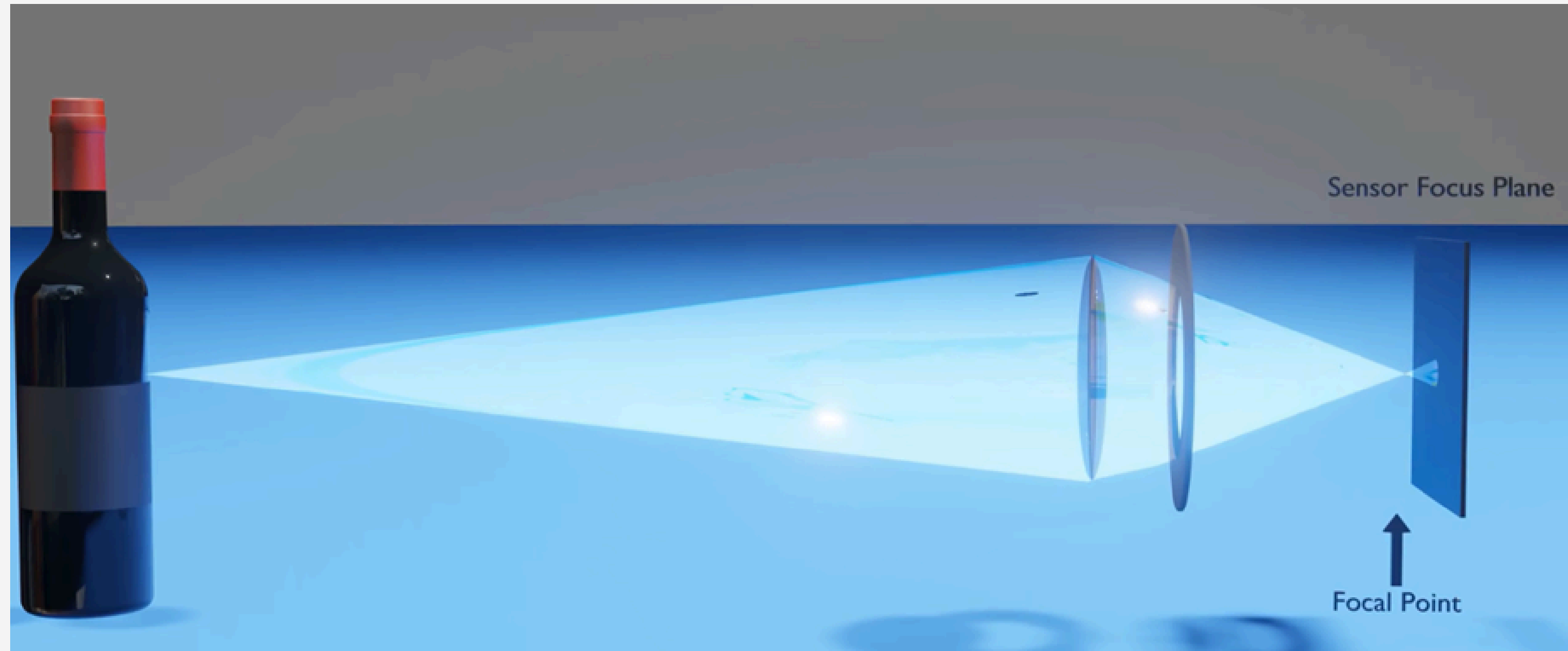
DEPTH OF FIELD

- The light passes through the lens, the cone of light reaching the sensor becomes larger.
- The focus of the light narrows down to a smaller point at the sensor focus plane, ensuring that the object (the Corkscrew) appears sharp.
- The sensor accurately captures the focal point, forming a focused image, indicating that the object is within the depth of field.

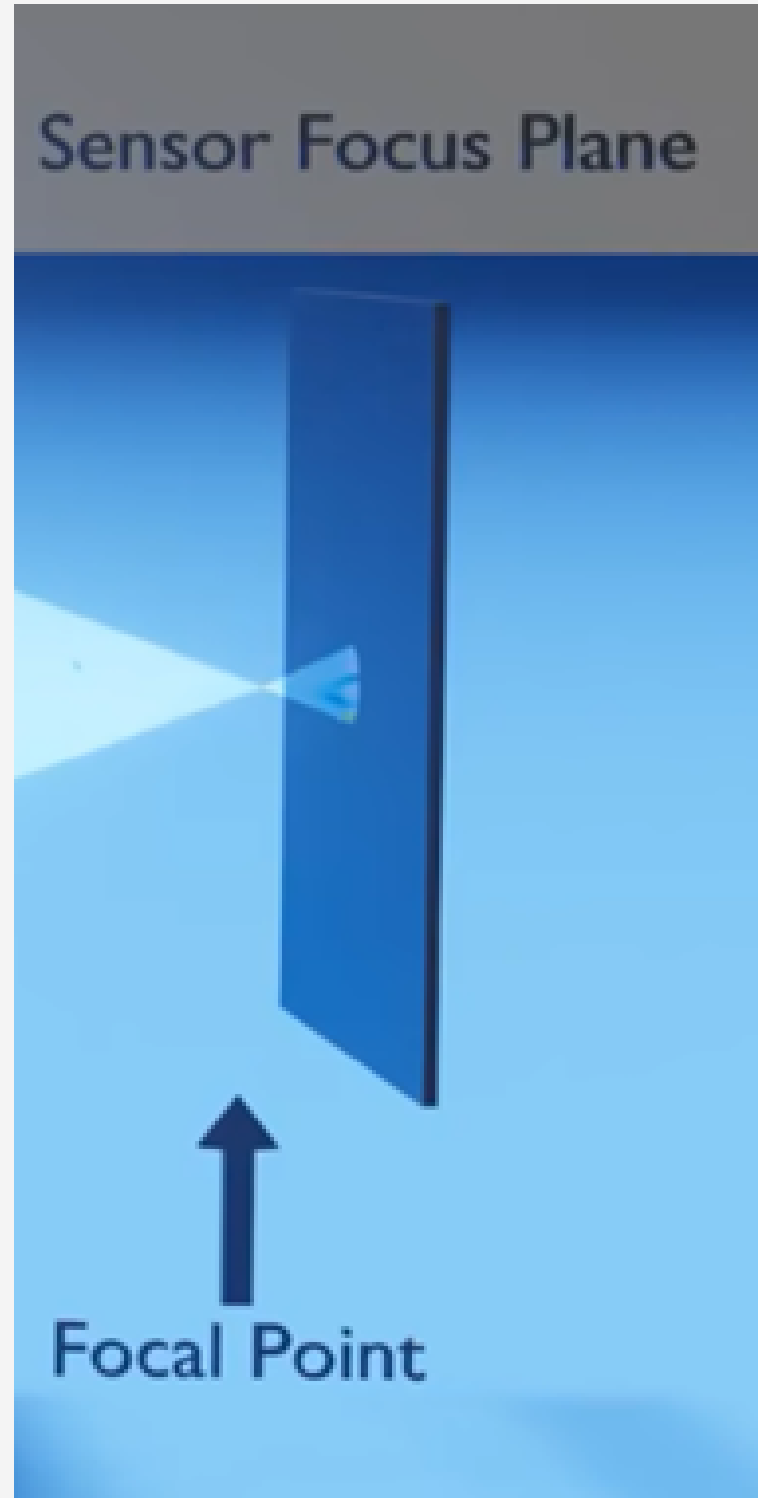


DEPTH OF FIELD

- The bottle is positioned farther from the sensor focus plane.
- As light passes through the focal point, it diverges, leading to defocus.



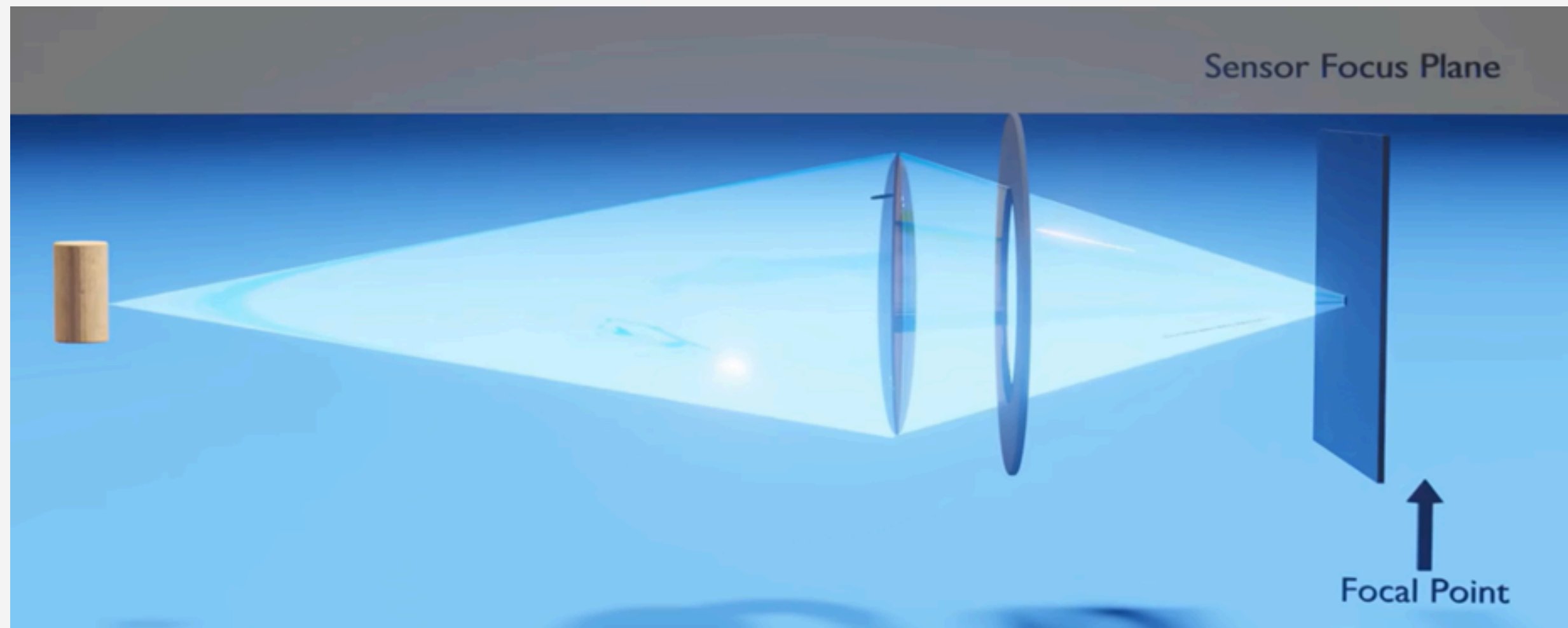
DEPTH OF FIELD



- The sensor reads the diverging light at a point.
- A larger light cone on the sensor creates a magnified and blurred image.
- The greater the divergence of light cones, the more severe the defocus.
- This illustrates that objects outside the depth of field will appear out of focus.

DEPTH OF FIELD

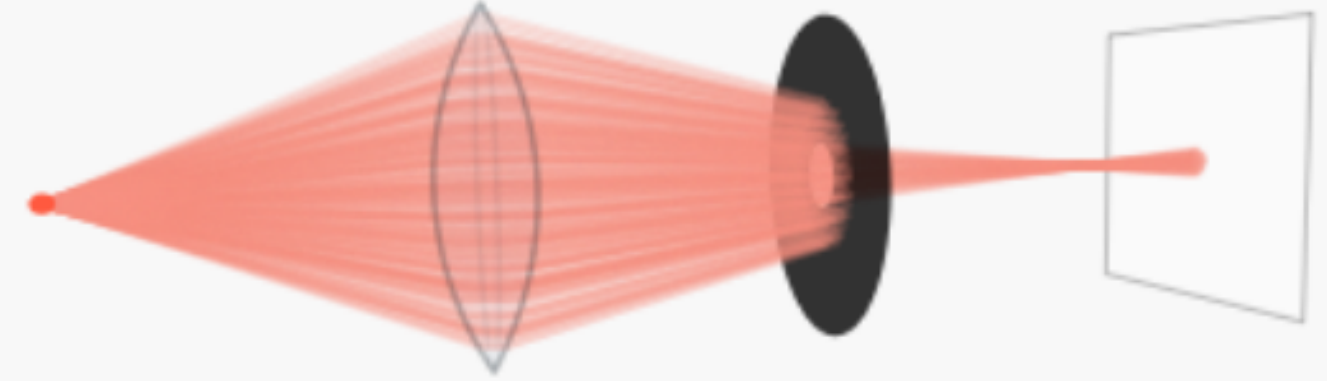
- The cork is positioned too close, causing the light rays to not converge at the focal plane before reaching the sensor.
- The sensor reads unfocused light rays, resulting in every point on the cork appearing blurred.
- Indicating it is outside the DOF range.



DEPTH OF FIELD



APERTURE SIZE



- Role of the Aperture
 - Introducing a stop with a hole, selectively allows certain rays of light to pass through.
 - The primary function of the aperture is to limit the angle of light passing through, narrowing the light cone.
- Adjusting the Light Cone Angle
 - Blocking Outer Rays:
 - The size of the aperture determines the range of light that can pass through.
 - Smaller apertures block more of the outer rays, resulting in a narrower light cone.
 - Reducing the Size of Blur Circles:
 - By restricting peripheral light rays, the circles of confusion on the sensor become smaller.

This reduces the blur effect in both the foreground and background, creating a sharper image.

EXPOSURE TRIANGLE

SPEED SHUTTER



APERTURE

ISO

WHAT IS EXPOSURE TRIANGLE ?

ANSWER

“Exposure Triangle” is showing 3 main factors of standard camera adjustment following lists below :

1

APERTURE

2

SPEED SHUTTER

3

ISO

APERTURE



1

Control volume of light that pass into camera sensor.

2

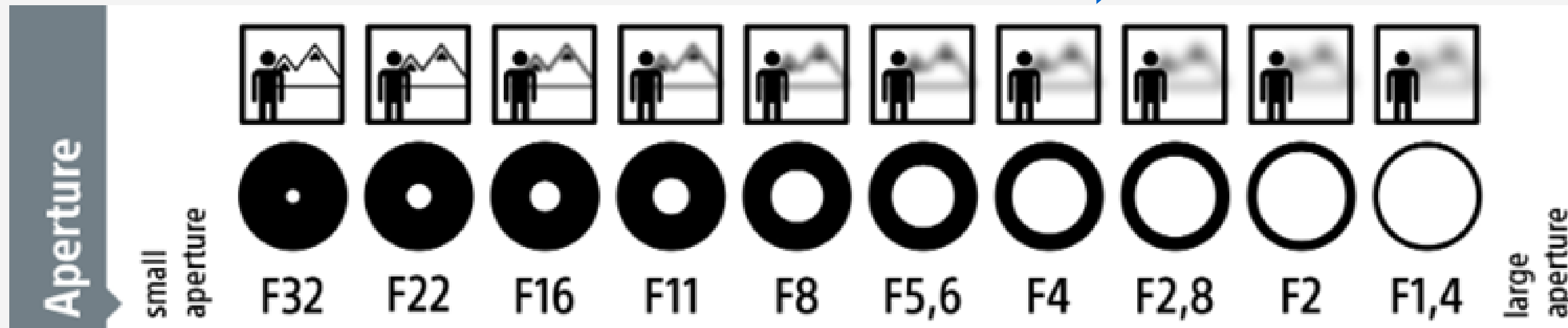
Control sharpness of both object and background.

3

Level size of aperture value is called "F-stop"

APERTURE

APERTURE VS. FOCUS



Low F-stop

Large aperture



**keep both
object + background**

High F-stop

Small aperture



**keep near object
(portrait photo)**

EXAMPLE

APERTURE MANY LEVEL



F2.8

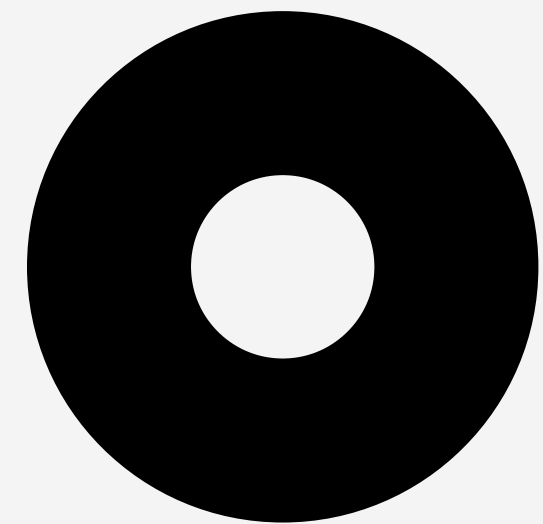
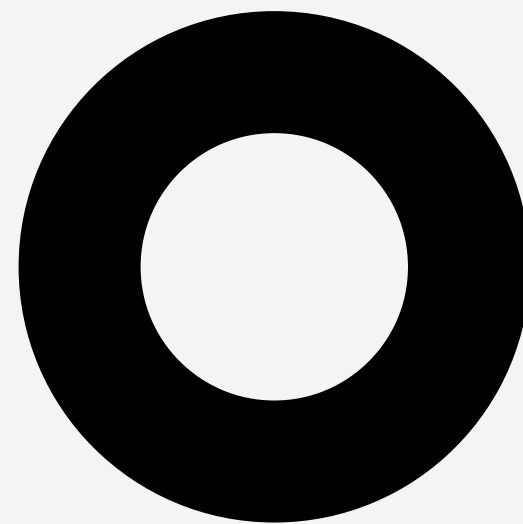
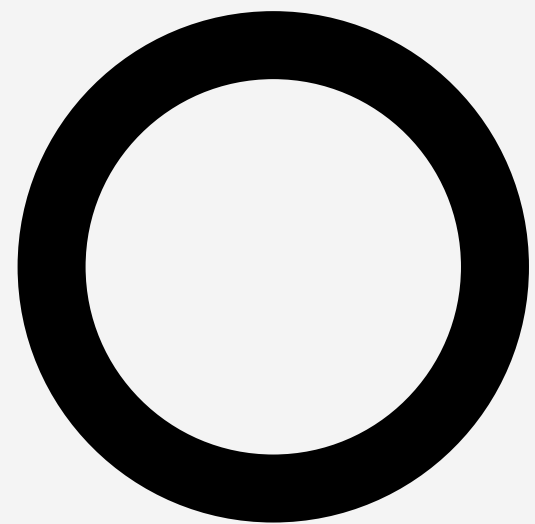


F4

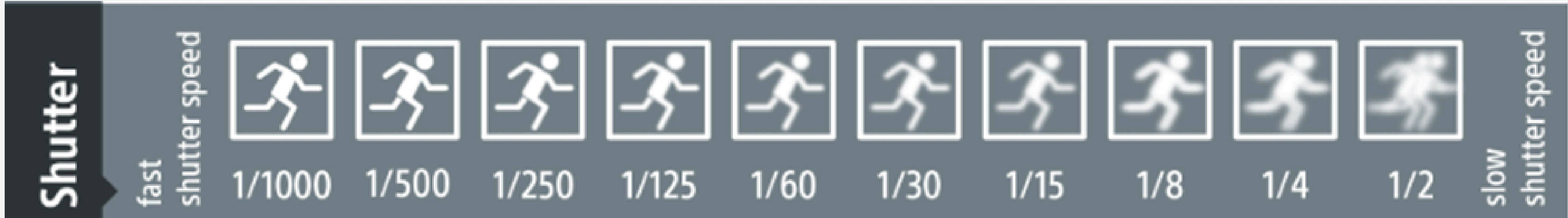


F5.6

SIZE OF APERTURE



- 



EXAMPLE

SPEED SHUTTER

Dark

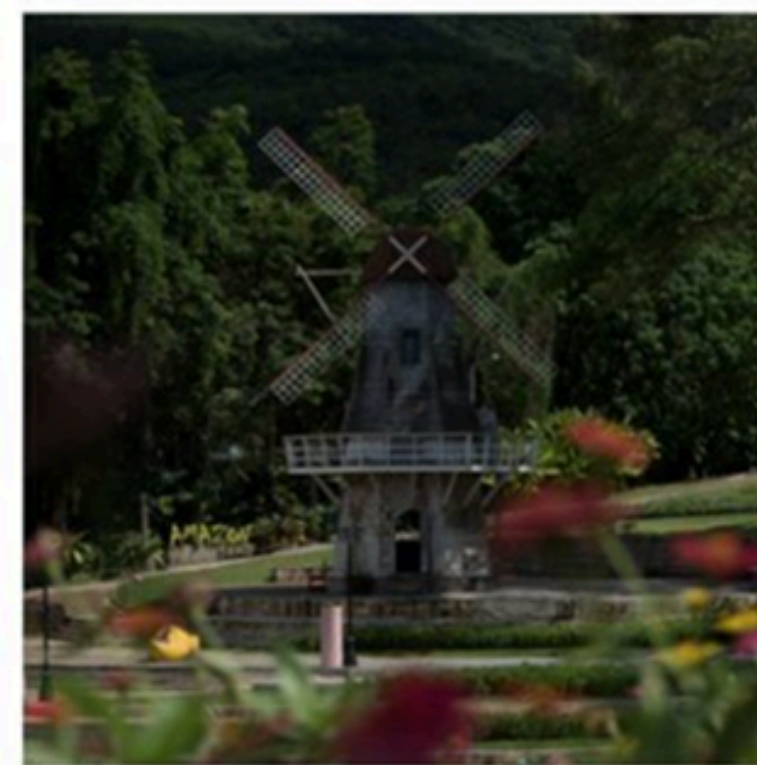
Fast speed shutter
can keep low volumn of light.



1/500



1/2000



1/8000

Slow speed shutter
can keep high volumn of light.

Bright

EXAMPLE

SPEED SHUTTER

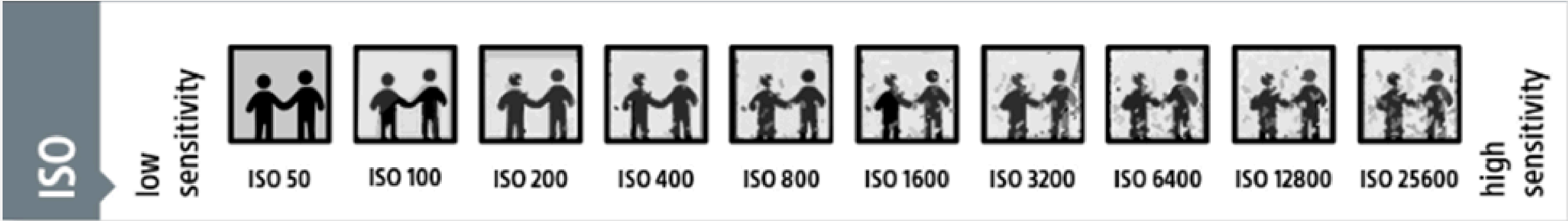
Slow speed shutter
can occur “Motions blur” .



Fast speed shutter
can snap moving object and give
stable photo result of object.



Warning! if set too high ISO,
our photo'll have noise.



THIS MEASURE CAN GIVE US
NORMAL BRIGHTNESS OF PHOTO.



YOUR TURN QUIZ !

Speed shutter = 1/1000

Aperture = F4

ISO = 400



If we adjust **Speed shutter = 1/500**
what's happen to our new photo ?

**THIS MEASURE CAN GIVE US
NORMAL BRIGHTNESS OF PHOTO.**

Speed shutter = 1/1000

Aperture = F4

ISO = 400



YOUR TURN QUIZ !

2

If we adjust **ISO = 100**
what's happen to our new photo ?

**THIS MEASURE CAN GIVE US
NORMAL BRIGHTNESS OF PHOTO.**

Speed shutter = 1/1000

Aperture = F4

ISO = 400

YOUR TURN QUIZ !

3

If we adjust **F-stop = 1.4**
what's happen to our new photo ?